



Romania's 2025 guide for health workforce attraction and retention: Bridging global strategies and local realities

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ABSTRACT

Background: Romania's health system faces a challenge of health workforce maldistribution, especially in rural and non-university urban areas. A national policy reform was launched in 2024–2025, supported by the National Recovery and Resilience Plan and WHO technical assistance. The reform's objective was to equip and empower local authorities to improve the recruitment, retention, and motivation of health professionals in their communities.

Reform content: The reform involves the development and nationwide dissemination of the "Solutions for Human Resources in Health: Guide of Local Actions". Developed through a participatory, evidence-based process with stakeholders, this guide is a practical toolkit for local authorities and hospital managers. It consolidates underused legal frameworks and curates national and international best practices examples across four main pillars: Education, Incentives, Regulation, and Support.

Expected Results: The primary expected outcome is an improvement in the geographical distribution of the health workforce, resulting in better-staffed facilities in underserved regions. A key secondary outcome is the boost in local authorities' administrative capacity and proactive involvement in the health workforce management, which fosters greater ownership and a more sustainable, decentralized approach to addressing staffing challenges. These outcomes will become measurable over the medium-to-long term as local authorities implement the guide's provisions.

Conclusions: The collaborative, evidence-based process used for the guide exemplifies a best practice model for stakeholder-driven policy development. The Romanian experience provides a transferable framework for countries with decentralized health systems to empower local actors in addressing health workforce maldistribution, though long-term success depends on local implementation and political will.

Research in Context

1. What is already known about the topic?

Health workforce maldistribution remains a global challenge, especially in rural and underserved areas. WHO and studies suggest retention needs multifaceted strategies, including financial

incentives, education, professional development, and social support. However, there is limited literature on how decentralized systems turn these principles into local policies.

2. What does this study add to the literature?

This manuscript provides an analysis of Romania's 2025 Guide of Local Actions, a reform aimed at tackling workforce maldistribution. It presents a detailed account of a collaborative, stakeholder-

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driven policy development process as a model of good practice. It also analyzes an innovative policy tool that focuses not on creating new national legislation, but on empowering local authorities by curating and clarifying existing legal instruments and showcasing national best practices.

3. What are the policy implications?

The Romanian model demonstrates that empowering local actors with well-structured information about existing policy levers is a powerful and efficient strategy for health system reform. For other countries with decentralized governance structures, developing similar implementation toolkits can bridge the gap between national strategy and local action. Effective, sustainable workforce policies should be co-created through participatory, evidence-based processes that ensure local relevance and ownership.

1. Background

The health and care workforce crisis is a defining challenge for health systems across the WHO European Region, threatening service sustainability and equitable access to care [1]. Romania's health system provides a compelling case study of this challenge, which has evolved significantly over the last decade [2]. For years, the primary issue was a severe "brain drain" of physicians and nurses migrating to Western European countries, creating a generalized workforce shortage [3,4]. In response, the government implemented a series of substantial, centrally-mandated salary increases beginning in 2018, as well as investments in healthcare infrastructure and increasing training capacity. This intervention proved highly successful in its primary goal, significantly curbing outward migration [2]. However, this success inadvertently unmasked and exacerbated a more complex, underlying challenge: the retained workforce became increasingly concentrated in major urban university centers, transforming a national shortage into a geographical maldistribution [2,5]. It is important to note that this internal maldistribution is not a recent phenomenon. It has been a long-standing challenge, historically driven by factors including the privatization of health services after 1989 and persistent internal migration from rural areas to larger cities. This has resulted in the emergence of "medical deserts", areas where population healthcare needs are unmet due to barriers like insufficient health professionals, lack of facilities, long waiting times, or disproportionately high costs [6]. The tangible impact of these medical deserts on the population is severe, particularly in specialized services, where patients in underserved regions face inadequate resources, prolonged waiting times for diagnosis, and significant out-of-pocket costs for travel, leading to poorer health outcomes [7].

Prior to the 2024–2025 reform, other fragmented strategies targeting the four pillars had been attempted with limited success. In terms of Education and Regulation, a "residency by vacancy" program was initiated in 2008 to place trainees in underserved hospitals, but it suffered from poor management and enforcement, with a high percentage of residents failing to fulfill their contractual obligations [4]. For Incentives, legal frameworks already permitted bonuses of up to 20 % for remote areas [8] and additional payments for difficult working conditions in primary care. Similarly, support mechanisms, such as provisions for rental housing for young specialists via the National Housing Agency and the ability for local authorities to rent spaces for medical practices, were also legally possible [9,10]. However, these tools were often implemented inconsistently, poorly funded at the local level, or unknown to local authorities who could use them. This fragmented and passive approach was insufficient to counter the complex drivers of maldistribution, creating the policy gap that the new "Guide of Local Actions" was designed to address.

This new understanding of the problem was progressively integrated into Romania's core strategic documents. The Multi-annual Strategy for

Human Resources for Health (HRH) (2022–2030) was the first significant policy document to place HRH governance and planning at the center of the national agenda, recognizing the need for a more sophisticated approach beyond simple salary incentives [11]. This narrative was further solidified in the National Health Strategy (2023–2030), which formally identified maldistribution as the key workforce issue to be addressed [12,13]. The impetus to act decisively gained momentum through a confluence of national and international policy developments. The European Union's National Recovery and Resilience Plan (PNRR) provided an unprecedented policy window, making substantial funding available for health system reforms contingent on addressing key structural weaknesses, with HRH development being a central pillar [11, 14]. This national effort is fully aligned with the broader European agenda. The 2023 Bucharest Declaration on the Health and Care Workforce symbolized this alignment and highlighted Romania's active role in regional policy dialogues [15,16]. Furthermore, the WHO Regional Committee for Europe adopted a comprehensive Framework for Action and a corresponding Resolution in 2023, urging member states to develop and implement context-specific strategies to improve workforce retention and distribution [17,18]. The "Solutions for Human Resources in Health: Guide of Local Actions" was therefore conceived as a direct, evidence-based response to this call, designed to equip and empower local actors within Romania's decentralized system and translate high-level national and international strategies and recommendations into concrete, local action [19,20].

2. Health policy processes

The development of the "Solutions for Human Resources in Health: Guide of Local Actions" was carried out through a multi-stakeholder consultative process from July 2024 to April 2025, led by the Ministry of Health (MoH) with technical assistance from the WHO, and was informed by engagement with national and local stakeholders through workshops and field visits. This engagement was facilitated by the NGO Doctors for Romania (Medici pentru România), which was contracted by the WHO at the request of the MoH. This process, undertaken within the framework of Romania's PNRR, was designed to ensure the final recommendations were grounded in both international best practice and the lived realities of Romania's underserved regions [19]. This initiative was a direct response to a recognized policy gap: the need to shift from addressing a general, national-level crisis of health professionals to tackling a more nuanced challenge of their geographical maldistribution. The process was designed to be collaborative and evidence-based, ensuring the final guide was grounded in both local realities and international best practices.

The process began with a comprehensive evidence-gathering phase. This included a systematic review of existing Romanian legislation to identify all current, yet often underutilized, legal mechanisms available to local authorities for attracting and incentivizing health workers. Concurrently, a desk review of international peer-reviewed and grey literature was conducted to identify global good practices in the recruitment, retention, and repatriation of health workers in underserved areas.

This evidence base informed a structured stakeholder engagement strategy. The selection of stakeholders was purposive, designed to convene a representative group of all key institutional categories involved in health workforce policy, finance, and service delivery. An initial workshop was held in Bucharest on July 5, 2024, convening approximately 40 key national stakeholders, including representatives from the MoH, the National Health Insurance House, hospital managers, professional organizations, and patient associations. This event served to present the initial findings from the desk reviews and to set priorities for the next phase of research. Following this, a series of seven field visits was conducted in July 2024 to hospitals in both underserved and well-staffed regions across four counties. These visits involved semi-structured interviews with hospital managers, physicians, and local

authority representatives (including municipal mayors) to collect qualitative data on existing challenges, successful local initiatives, and the perceived feasibility of various incentive models.

The insights from both the evidence reviews and the fieldwork were synthesized and presented at a second, more intensive workshop in Azuga on September 20–21, 2024. This event brought stakeholders together to discuss the findings and collaboratively agree on the core structure of the guide. Through this participatory process, the four key pillars of the guide, Education, Incentives, Regulation, and Support, were agreed, ensuring the final document directly addressed the needs and priorities identified by those working on the ground.

The final drafting was undertaken by a team of national experts from the MoH, with technical support from WHO. A draft version of the guide was publicly launched for final consultation on April 15, 2025, allowing for feedback from a wider audience, including health managers, academic leaders, professional bodies, patient representatives, and local leaders, before its official publication and dissemination. This systematic, transparent, and stakeholder-driven process was designed not only to produce a practical policy tool but also to build consensus and ownership among the local actors responsible for its implementation. The key phases of this multi-stage policy process are summarized in Fig. 1. The collaborative nature of this reform involved a diverse set of actors, whose primary roles and relationships are depicted in the stakeholder ecosystem map in Fig. 2.

3. Content of the reform

The “Solutions for Human Resources in Health: Guide of Local Actions” is a strategic policy instrument designed to translate national health objectives into practical, localized interventions [19]. It operationalizes the principle that while the central government sets the overarching strategy, successful implementation in a decentralized

system depends on empowering local actors. The guide’s content is structured around four interconnected pillars, aligning with the World Health Organization’s (WHO) global recommendations for workforce development in rural and underserved areas: (1) Education, (2) Incentives, (3) Regulation, and (4) Support [19,21]. An overview of the core interventions within each pillar, alongside the international evidence that informed them, is presented in Table 1. The following sections provide a more detailed description of these pillars.

The educational pillar aims to create a sustainable pipeline of professionals predisposed to practice outside major urban centers, a strategy of global relevance. The guide recommends collaborations between local authorities and medical universities to recruit students from rural backgrounds [19]. This reflects a broad international consensus that a rural origin is one of the strongest predictors of a physician ultimately choosing to practice in a similar setting, with evidence from systematic reviews covering low, middle, and high-income countries [22–25]. An innovative recommendation is the accreditation of non-university hospitals as training centers for medical residencies and other categories of professionals [19]. This marks a significant departure from the traditional university-centric model and directly applies lessons from countries like Australia and Canada, where longitudinal rural training has proven effective for retention [23,26]. By creating local academic career pathways and funding continuous professional development, the guide directly addresses the key “push” factor of professional isolation, a challenge frequently cited in the literature [27,28].

To make regional posts more attractive, the guide outlines a package of incentives that can be deployed at the local level. It empowers municipalities by clarifying their legal authority under existing laws (e.g., Law 95/2006, Law 153/2017) to offer financial bonuses, housing subsidies, and transport allowances [8,19]. This approach is consistent with a large body of international evidence, which indicates that while financial attributes are essential, they are rarely sufficient on their own

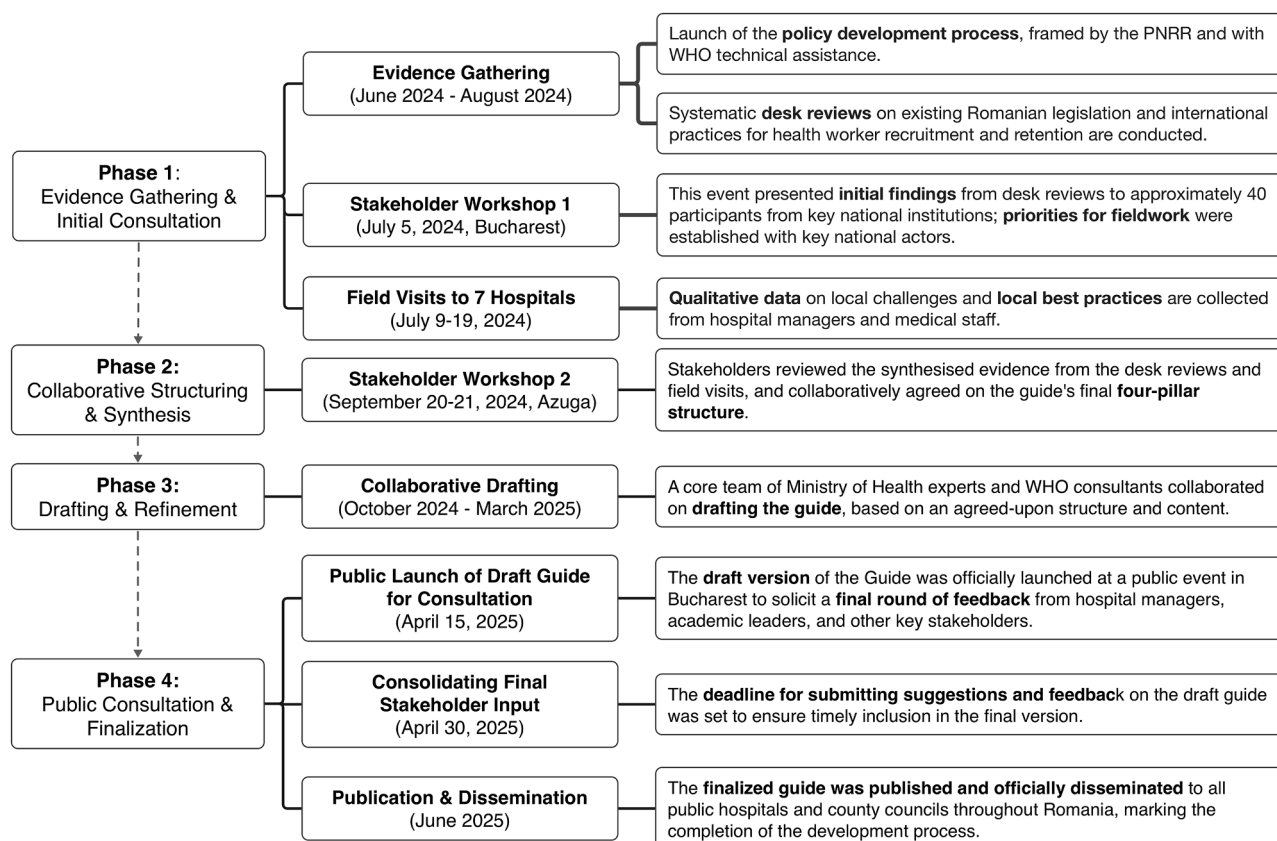


Fig. 1. Timeline of the Guide's development process.

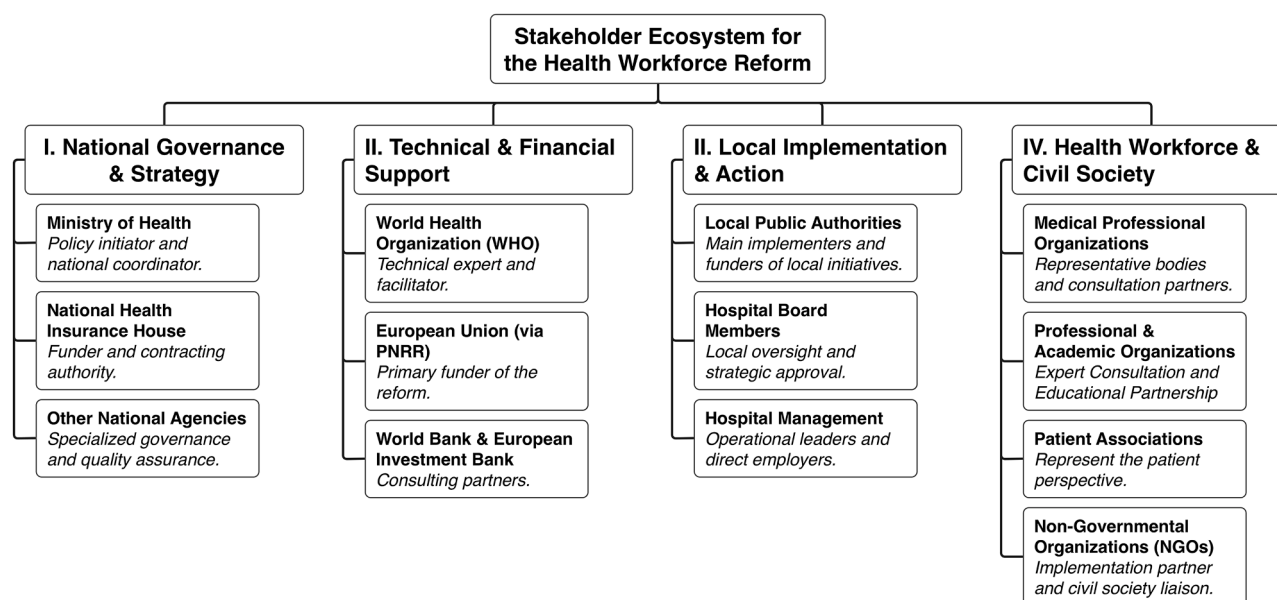


Fig. 2. Stakeholder ecosystem for the health workforce reform.

[24,25]. Effective strategies globally rely on a bundled package of interventions. Discrete choice experiments in countries like Portugal and the Philippines have demonstrated that physicians value non-financial attributes, such as professional supervision and good working conditions, alongside salary [29,30]. The guide's inclusion of a forthcoming performance-based pay mechanism also reflects an international trend towards rewarding quality and efficiency [19]. However, a key challenge for this reform is that its success is contingent on the highly variable financial capacity of Romania's local administrative units. The guide itself does not allocate new central funds, thus creating a potential risk for further regional disparities if wealthier municipalities are better able to fund more attractive packages. Furthermore, beyond the financial aspect, the overall success of the guide's recommendations will require a significant investment of time and administrative resources. It will ultimately depend on the sustained political will of local authorities to implement them.

The core logic of the reform is captured in its regulatory and support pillars. The guide functions primarily as an instrument of empowerment, clarifying for local leaders the scope of their existing authority to invest in and manage local health services [19]. This decentralized model is a pragmatic response to the chronic challenge of maldistribution and mirrors strategies employed by other European countries, such as Sweden, which have leveraged regional-level planning to address geographical inequities [31]. The support pillar complements this by focusing on tangible improvements to the work environment and, crucially, integrating the families of relocated professionals into the community [19]. This holistic approach, addressing spouse employment and children's schooling, is recognized in extensive international qualitative research as a critical factor for long-term retention [28]. Studies consistently show that family and community integration are often decisive in a physician's decision to remain in a rural post, making this pillar a key strength of the guide's design [30,32,33].

4. Expected outcomes

As the "Solutions for Human Resources in Health: Guide of Local Actions" was launched in April 2025, empirical outcome data is not yet available [19]. However, the guide's objectives and the international evidence on which its recommendations are based allow for a clear articulation of its expected outcomes, success indicators, and potential challenges. The primary expected outcome is a tangible reduction in the

chronic maldistribution of the health workforce in Romania, thereby improving equity in access to healthcare for populations in rural and non-university urban areas [19,27]. Success will likely be evaluated against a composite of specific, measurable indicators benchmarked against international experiences. These indicators include a quantifiable increase in the physician-to-population ratio in targeted underserved areas, mirroring the metrics used to evaluate Brazil's successful "Mais Médicos" programme, which doubled the number of municipalities with at least 1.0 physician per 1000 inhabitants [35]. Another key metric can be an increase in the 3- and 5-year retention rates for newly recruited physicians in these areas, a standard measure of long-term success used to validate targeted educational programs like the CPIRD initiative in Thailand, which demonstrated a 2.4-fold higher likelihood of 3-year retention for its graduates [34]. In the longer term, these workforce improvements are expected to translate into better population health outcomes, such as a reduction in avoidable hospitalizations, an indicator also successfully employed in the Brazilian evaluation [35].

The reform incorporates mechanisms for monitoring and evaluation, signaling an intent for data-driven policy adjustment. The guide itself is explicitly presented as a "living document," with an open invitation for stakeholder feedback to inform future iterations, suggesting an adaptive policy cycle rather than a static intervention [19]. This approach is reinforced by the guide's own emphasis on systematic monitoring as a key condition for success [19]. In this decentralized model, implementation in Romania is led by the local stakeholders, primarily local public authorities and hospital managers, who are the guide's primary audience [19]. The "early adopters" are effectively the sources of the national best practices curated within the guide, such as the successful retention strategies observed in Bistrița and Buhuși, which now serve as documented models for other communities [19]. The guide's explicit nature as an evolving "living document" that actively solicits stakeholder feedback confirms the commitment to this ongoing, adaptive monitoring process.

Despite its evidence-informed design, the reform faces potential challenges. International experience offers cautionary tales that are particularly relevant to the Romanian context. In Scotland, a significant financial bonus offered through the Targeted Enhanced Recruitment Scheme was found to have only a marginal influence on trainees' location choices, with only 21 % of participants influenced by the payment [41]. Furthermore, an evaluation of Australia's General Practice Rural Incentive Program showed that while it increased the number of

Table 1

A summary of the guide's key recommendations across the four pillars, cross-referenced with supporting international practices.

Pillar & Recommendations	Supporting International Practice & Evidence	Summary of the Action Plan	Expected Outcomes
I. EDUCATION			
Targeted Recruitment Campaign	Systematic reviews confirm rural origin and rural clinical training are strong predictors of rural practice across diverse national contexts [22–24]. Interventional studies in Thailand demonstrate the success of such targeted educational programs [34].	The residency program expands to County Hospitals, and universities of medicine and pharmacy extend nursing programs to local communities. A “medical scouting” strategy allows smaller hospital boards to connect with major university centers for final-year residents. Simultaneously, integrated employment policies recruit specialists from various fields to build effective multidisciplinary teams.	- Increased pipeline of medical graduates choosing to practice in their home regions. - Stronger professional ties between new doctors and local communities, leading to higher initial recruitment.
Supporting Continuous Medical Education and professional development activities	The need for constant education, information updates and training is widely recognised. Across different regions of the globe, the problem of reduced medical education within the practitioners from small towns/rural areas is risen.	The development of locally funded study grant programs is necessary to alleviate the financial burden of Continuing Medical Education (CME) for physicians practicing in underdeveloped regions.	- Alleviating clinician isolation in small communities and expanding local healthcare services reduces travel for care and helps attract and retain physicians in underserved areas.
II. INCENTIVES			
Bundled incentive packages with bonuses, housing subsidies, and transport allowances funded by local authorities.	The literature consensus is that financial-only incentives are insufficient [24, 25]. Effective strategies, as seen in Brazil and France, bundle financial rewards with professional and personal support [35,36].	Financial incentives like bonuses, service housing, or rent subsidies help newly employed physicians integrate into the community.	- Higher uptake of vacant posts in underserved areas. - Improved short-to-medium term retention by addressing immediate financial and logistical barriers to relocation.
III. REGULATION			
Empowering the Local Communities	Decentralized approaches are increasingly used to address regional disparities. Sweden, for instance, uses regional-level	Dissemination of current statutory regulations to local authorities (County and Local Councils) concerning the permissible bonuses that may	- Development of flexible and locally owned retention strategies tailored to specific community needs.

Table 1 (continued)

Pillar & Recommendations	Supporting International Practice & Evidence	Summary of the Action Plan	Expected Outcomes
Recognition of International Specialists	planning to counterbalance geographical imbalances in its health workforce [28].	be allocated to subordinate medical personnel.	- Increased accountability of local government for local health system performance - An increasing inflow of foreign medical professionals is choosing to integrate into the Romanian healthcare system
	Due to socio-economic causes a lot of Romania medical professional choose to emigrate in the past years, creating a significant diaspora [37,38].	The medical diaspora is a valuable source of human resources. To encourage these practitioners to return to Romania, procedures for professional recognition need to be simplified and decentralized.	
IV. SUPPORT			
Infrastructure Development	Empirical evidence from Eastern European nations, including Poland [39] and Romania [40], substantiates this premise, demonstrating that inadequate infrastructure accelerates the out-migration of medical professionals toward better-resourced regions.	Strategic investment in healthcare capital assets: advanced medical technology, facility renovation, and new hospitals.	- Decelerate the emigration of healthcare professionals from areas where new hospitals and modernization projects are implemented.
Family Integration	Qualitative studies from high-income countries emphasize that community fit, spousal career opportunities, and family well-being are critical determinants of a physician's decision to remain in a rural area long-term [24,29,30].	Since integrating a healthcare professional into a hospital also implies including their family in the community, the guide suggests creating integrated plans involving local authorities and businesses. These plans aim to provide competitive jobs for physicians' partners.	- Reduced personal and family-related barriers to relocation. Significantly improved long-term retention rates of healthcare workers and their families in the community.

physicians, it produced weak evidence of reducing patient waiting times, suggesting that simply boosting supply does not automatically resolve complex access-to-care barriers [42]. A key precaution for Romanian policymakers will therefore be to closely monitor the equity of the guide's implementation and consider targeted national-level support mechanisms for the most disadvantaged regions, ensuring the reform narrows, rather than widens, the country's profound health disparities.

5. Conclusion

The 2025 Guide of Local Actions represents a significant evolution in Romania's approach to health workforce policy. More than just a

technical document, it serves as a compelling case study in collaborative, evidence-based, and stakeholder-driven policy development. Its primary strength lies not in creating new legislation, but in strategically empowering local actors through the consolidation and clarification of existing legal frameworks. By moving away from a rigid, top-down mandate and instead offering a flexible toolkit of curated best practices, the reform provides a nuanced response to the complex challenge of health worker maldistribution within a decentralized system.

The participatory process used to develop the guide, combining international evidence reviews with extensive national and local stakeholder consultations, was crucial for ensuring its relevance and fostering a sense of ownership among those responsible for its implementation. This collaborative model is a notable practice, particularly for health systems in Central and Eastern Europe seeking to build consensus and bridge the gap between national policy objectives and local realities.

However, the guide's success will ultimately depend on the sustained political will and administrative capacity of local authorities to utilize the tools it provides. Nonetheless, this reform is a critical and well-designed first step. By focusing on empowering local communities with the knowledge and legal clarity needed to create attractive professional and social environments, the Romanian experience offers a valuable lesson for other countries: that building local capacity and fostering collaborative governance can be one of the most effective strategies for solving entrenched health system challenges.

Consent for publication

Not applicable.

Ethics approval and consent to participate

The study received ethical approval for the secondary analysis of policy documents from the Research Ethics Subcommittee of the “Carol Davila” University of Medicine and Pharmacy, Bucharest (Approval No 24,373/23.09.2025). The workshops and interviews that informed the manuscript's findings were conducted as a policy-development exercise with professional stakeholders and did not constitute human-subjects research.

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Data sharing statement

This manuscript is based on a secondary analysis of policy documents and materials generated during the 2024–2025 national health workforce reform process in Romania. The data analyzed herein consist of: (1) the final, publicly available “Solutions for Human Resources in Health: Guide of Local Actions” policy document, and (2) synthesized reports and proceedings from the stakeholder workshops that informed its development. The raw qualitative data, collected primarily for policy development purposes, including individual interview transcripts and facilitators' notes, are not publicly available to protect participant confidentiality. Inquiries regarding the publicly available source materials and synthesized reports can be directed to the corresponding author upon reasonable request.

CRedit authorship contribution statement

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Declaration of competing interest

The authors declare that they have no conflicts of interest. The views expressed herein are those of the authors and do not necessarily represent the decisions or the stated policy of the World Health Organization or the Romanian Ministry of Health.

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